Tester Present

Specialist Automotive Solutions

TESTER ENGINEERING SUITE · V1.0.9-2026

TESTER ENGINEERING SUITE

PRODUCT BROCHURE

J2534 ECU diagnostics · flash · UDS · live datalogging · calibration

"Engineered like infrastructure, not a scan tool."

PLATFORM SAE J2534 PassThru · UDS ISO 14229-1
SYSTEMS Windows 7+ (x86 / x64)
WEB tester.engineering



Right to Repair

BUILT BY A MECHANIC, FOR THE RIGHT TO REPAIR

THE COMPLETE DIAGNOSTIC & FLASH PLATFORM

ONE PLATFORM, FROM THE OBD PORT TO THE BENCH

Tester Engineering Suite is a professional-grade Windows platform that unifies J2534 ECU diagnostics, bench programming, live datalogging, USB oscilloscope capture, CAN-bus investigation and advanced vehicle configuration — for technicians, calibrators and OEM engineering teams who need real coverage without a dealer-only gate or a subscription that switches off when the signal drops.

Built on the industry-standard SAE J2534 PassThru API, it works with any compatible pass-through interface and scales from a quick fault read on the driveway to a full flash session on the bench. Every workspace is built to the same standard: explicit about what it can and cannot do, validated against real vehicles, and honest about risk.

700,000+	90,000+	20,000+	100+	60+
OEM FAULT CODES	DIAGNOSTIC ROUTINES	PID FORMULAS	SEED-KEY ALGORITHMS	OEM FAMILIES

TIP **Right to repair, by design**

Built by a trade-qualified mechanic for independent workshops and engineers. No automatic subscriptions, no per-feature paywalls, no background billing — a hardware-fingerprinted licence you own.

01 CAPABILITIES

Everything below ships in the box. Feature availability follows the connected interface and the vehicle.

DIAGNOSTICS & CONNECTIVITY

- **General OBD-II —** SAE OBD-II Modes 01–0A: live PIDs, freeze-frame, stored/pending/permanent DTCs, readiness monitors, on-board test results and vehicle information.
- **Full UDS (ISO 14229-1) —** session control, security access, read/write DID, routine control, I/O control, memory read/write, ECU reset and communication control across a comprehensive service surface.
- **VIN read & decoder —** one-touch VIN read with a full WMI / VDS / VIS breakdown and vehicle attribution.
- **Fault-code management —** global and OBD-II code handling with freeze-frame capture, severity hints and guided triage.
- **CAN & K-Line bus topographer —** active bus discovery that maps which modules answer on which bus, with safe read-only probing.
- **Multi-bus CAN sniffer —** HS / MS / SW-CAN capture with filtering, replay and per-ID annotation.
- **CAN bit-timing analyser —** diagnose flaky buses: termination, sample-point and bit-timing checks per SAE J2284-3.
- **DoIP & CAN-FD ready —** modern transports with automatic detection behind the same diagnostic surface.

LIVE DATA & ANALYSIS

- **1000+ PID definitions —** OEM and generic PIDs with scaling and units, plus a custom PID builder for anything not in the library.
- **Live datalogger & signal analyser —** multi-channel waveform plotting, dashboards, playback and histogram analysis.
- **USB oscilloscope integration —** PicoScope, SCPI/LXI bench scopes and Hantek 6022 captured alongside J2534 data on one timeline.
- **Relative compression & misfire —** cylinder contribution and misfire detection from RPM ripple, no engine teardown.
- **Acoustic diagnostics & NVH —** FFT spectrogram and order-tracking cursors for chasing noise, vibration and harshness.

PROGRAMMING & FLASH

- **ECU reflashing —** validated flash profiles across multiple vehicle families, with dry-run validation before any write.
- **Seed-key security —** 100+ security-access algorithms including UDS $27 families, covering the modules that gate reprogramming.
- **Guided service routines —** one-click routine runner for configuration, resets and service functions, with plain-language descriptions.
- **As-built / configuration editor —** read, edit and write module configuration blocks with change tracking.

- **Immobiliser & key workflows —** guided anti-theft / key-programming wizards with live security-state monitoring.
- **ODX → routine importer —** turn standardised diagnostic descriptions into runnable routines.

> **WARNING** **Safety is not optional**
>
> Restraint and airbag modules are hard-blocked from routine and output-state commands. Emissions-relevant edits are double-confirmed. Battery voltage is supervised throughout every flash, with transaction-level abort and post-flash calibration-ID / CVN verification.

CALIBRATION & MEMORY

- **A2L / XDF calibration editor —** ASAM MCD-2 MC (A2L) and TunerPro XDF map editing with live heatmap, 3-D surface view and map math.
- **Checksum correction & safe write-back —** CRC / Fletcher checksum handling with verified, reversible write-back.
- **EEPROM & cluster tooling —** bench EEPROM editor and module tooling spanning thousands of vehicle variants — strictly offline, never on the vehicle bus.
- **Cluster graphics & flash —** instrument-cluster graphics and flash tooling for supported clusters.
- **Crash-data extraction —** read restraint / airbag event records (EDR/CDR) and produce a signed, tamper-evident report.

SPECIALTY & BENCH TOOLS

- **IM240 / RG240 emissions lane —** transient-cycle emissions test lane with compliant mass-emission calculation.
- **File & binary utilities —** VBF unpack/repack, hex viewer, checksum calculators and format converters (S19 / Intel HEX / raw BIN).
- **Bluetooth diagnostic suite —** structured Bluetooth assessment workflow, plus a hidden-BLE tracker sweep for Apple Find My, Tile, Samsung and Google tags.
- **Remote master / slave agent —** drive a shop bench from another site over an encrypted WebSocket link, with full feature parity.
- **Advanced & developer menus —** Ctrl+Shift+A opens 30+ engineering tools; Ctrl+Shift+E exposes deep suite diagnostics (see the Power-User Addendum).

SHOP WORKFLOW & RECORDS

- **VIN intelligence brief —** registration, recall and complaint intelligence from Australian and NHTSA sources, explicit about what a plate can and cannot feed.
- **Per-VIN technician notes —** supersede-not-delete history so nothing is ever silently overwritten.
- **Service schedule & maintenance —** interval library and maintenance planning per vehicle.
- **Work orders & reporting —** labour timer, work orders and PDF reports with tamper-evident manifests.
- **Pre / post scan & DVI —** insurer-ready before/after scans and digital vehicle inspection.

02 TESTER ENGINEERING HEAVY

Built for the pit and the paddock: fully offline, no phone-home, USB-key licence.

The heavy-duty edition extends the same platform to trucks, buses and mobile plant, with the datalinks and protocols the light-vehicle world never sees — and a hard rule: zero internet, ever.

- **SAE J1939 diagnostics —** DM1–DM19 fault handling, module discovery and live SPN/FMI decoding.
- **RP1210 adapter support —** TMC RP1210 interfaces from Nexiq, Dearborn, DrewLinQ and Jaltest.
- **Caterpillar Data Link (CDL) —** CAT CDL monitoring and diagnostics via supported adapters.
- **J1587 / J1708 legacy datalink —** pre-2007 heavy-vehicle diagnostics.
- **UDS-on-J1939 —** modern UDS services over J1939 for Isuzu, Hino, Fuso and Paccar-family modules.
- **Offline-first —** no phone-home, USB-key licence; everything works with the network cable pulled.

OEM COVERAGE AT A GLANCE

Coverage is validated per family and grows with every release. Anything outside a validated profile falls back to the generic SAE J2012 decode set (3,503 DTCs) so you are never left blind.

OEM FAMILY	DTCS	PID FORMULAS	DATA IDS (DIDS)
Ford	3,138	3,420	2,180
General Motors	1,802	2,310	1,605
VAG (VW/Audi/Škoda/SEAT)	1,431	3,180	1,872
BMW · Mercedes-Benz	validated	validated	validated
Toyota · Honda · Nissan	validated	validated	validated
Hyundai/Kia · Mazda · Subaru	validated	validated	validated
Jaguar/Land Rover · Volvo · Stellantis	validated	validated	validated
Generic SAE J2012 fallback	3,503	–	–

Bundled knowledge base: **17,974 OEM-specific DTCs · 660 Mode-06 test definitions · 60+ OEM families.**

03 PROTOCOLS, STANDARDS & HARDWARE

STANDARDS SUPPORTED

DOMAIN	STANDARDS
Diagnostic	UDS ISO 14229-1 · SAE J2534-1/2 · OBD-II · SAE J1979 / J1979-DA · J2012 · ISO 22900-2 · ODX-D
Transport	ISO 15765 (CAN) · ISO-TP · DoIP / ISO 13400 · CAN-FD · GMLAN · ISO 9141-2 · ISO 14230 KWP2000 · KWP1281 · SAE J1850 VPW/PWM
Heavy-duty	SAE J1939 (+ -21/-73/-81) · J1708 · J1587 · Caterpillar CDL
Calibration	A2L / ASAM MCD-2 MC · TunerPro XDF 1.6 · VBF · Motorola S19 · Intel HEX · raw BIN

INTERFACES

Any SAE J2534 PassThru interface is supported. Verified devices include OBDX Pro, Ford VCM2, Drew Technologies Mongoose / CarDAQ-Plus / M / X, Tactrix OpenPort, VXDIAG VCX and Bosch Mastertech. ELM327 is supported for basic OBD-II only — not for flash or security access.

> **NOTE**
> ELM327 is a convenience path for reading generic OBD-II. It cannot perform UDS security access or reprogramming; the suite will tell you when a task needs a real pass-through device.

04 LICENSING

You decide when — or if — you renew.

No automatic subscriptions, no background billing, no per-feature unlocks. Licences are bound to your machine fingerprint; lifetime tiers ship with a USB hardware token.

TIER	PRICE (AUD)	WHAT YOU GET
12-Month Licence	$800 / year	Full light-vehicle platform. Does not auto-renew — you choose to extend.
Lifetime Licence	$2,500 once	Full platform for life, priority support and a USB hardware token.
Tester Engineering Heavy	$4,500 once	Lifetime heavy-duty edition: J1939 / RP1210 / CDL / J1587, fully offline.

Companion — Tester Engineering Lite

A free Android app (signed APK, no store account required) paired with the OBDX Pro adapter: live data, DTC read/clear and performance tracking in your pocket.

> **TIP** **Request a licence**
>
> Talk to us about a licence, a technical evaluation or a custom solution: tester.engineering
>
> *Built by a mechanic, for the right to repair.*